

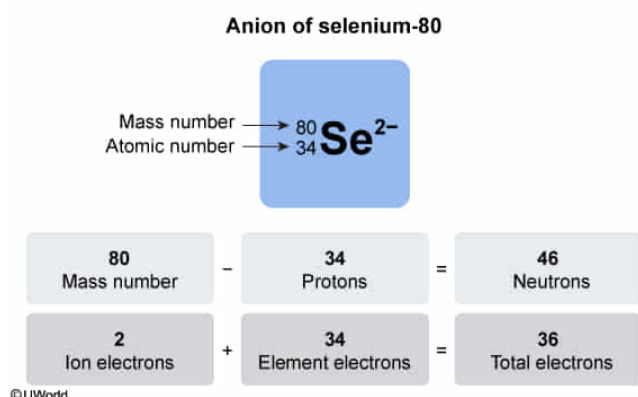
Atoms of a given element have a defined number of protons, neutrons, and electrons, but atoms can gain or lose electrons to form ions. How many neutrons and electrons are present in a $^{80}\text{Se}^{2-}$ ion formed when a neutral selenium-80 atom gains two electrons?

- ☐ A. 80 neutrons and 36 electrons [6%]
- ✓ ☒ B. 46 neutrons and 36 electrons [87%]
- ☐ C. 34 neutrons and 34 electrons [1%]
- ☐ D. 46 neutrons and 34 electrons [3%]

Correct

87% answered correctly

Explanation:



Isotopes are specified either by using a chemical symbol with the **mass number** given as a superscript to the upper left (eg, ^{80}Se) or by stating the element name hyphenated with its mass number (eg, selenium-80). Because the **atomic number** (the number of protons) determines the *identity* of an element and is the same for all atoms of that element, the atomic number is not always explicitly stated in **isotope notation**; however, when stated, it is placed as a subscript to the lower left of the chemical symbol (below the mass number).

An atom of any isotope in its elemental state will be **neutral** (have **no net charge**), and the number of negatively charged electrons will be equal to the number of positively charged protons. However, elements may react to form ions by gaining or losing electrons. A neutral atom will **acquire a net charge of +1 for each electron lost** and a net charge of **-1 for each electron gained**.

Selenium-80 has a mass number (protons plus neutrons) of 80. Because all atoms of selenium have 34 protons, the number of neutrons is found by subtracting the atomic number (protons) from the mass number ($80 - 34 = 46$). A neutral (uncharged) atom of selenium-80 in its elemental state has 34 protons and 34 electrons. However, the -2 charge in the $^{80}\text{Se}^{2-}$ ion indicates that the ^{80}Se atom has gained two *additional* electrons during ion formation. Therefore, a $^{80}\text{Se}^{2-}$ ion has 46 neutrons and 36 electrons.

(Choice A) The mass number (protons plus neutrons) is 80.

(Choices C and D) ^{80}Se has 34 protons, 46 neutrons, and 34 electrons, but $^{80}\text{Se}^{2-}$ has 34 protons, 46 neutrons, and 36 electrons.

Educational objective:

Isotopes are specified by their mass number (the number of protons and neutrons in the nucleus). The number of neutrons in an isotope can be determined by subtracting the element's atomic number (number of protons) from the mass number. An atom in its elemental state is neutral (equal number of protons and electrons) but acquires a net charge of $+1$ for each electron lost and -1 for each electron gained during ionization.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402586

System:
Atomic Theory and Chemical Composition

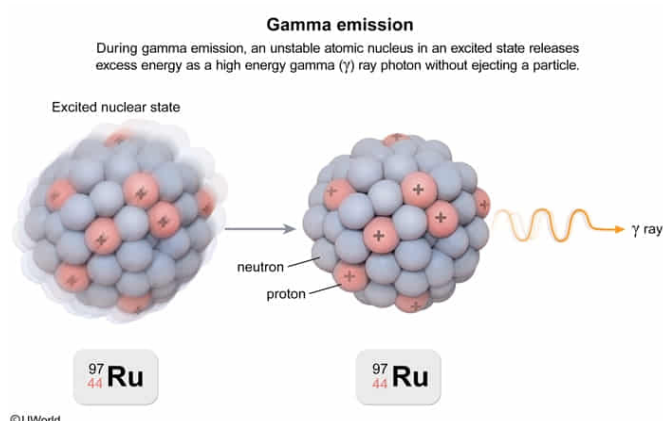
Which of the following equations correctly represents the gamma decay of an atom of ruthenium-97?

- ☐ A. ${}^{97}_{44}\text{Ru} \rightarrow {}^{97}_{45}\text{Rh} + {}^0_{-1}\text{e} + {}^0_0\bar{\nu}$ [3%]
- ☐ B. ${}^{97}_{44}\text{Ru} \rightarrow {}^{93}_{42}\text{Mo} + {}^4_2\text{He}$ [2%]
- ☐ C. ${}^{97}_{44}\text{Ru} \rightarrow {}^{97}_{43}\text{Tc} + {}^0_{+1}\text{e} + {}^0_0\nu$ [3%]
- ✔ ☒ D. ${}^{97}_{44}\text{Ru} \rightarrow {}^{97}_{44}\text{Ru} + \text{photon}$ [90%]

Correct

91% answered correctly

Explanation:



Radioactivity is the spontaneous emission of radiation (photons or particles) from an unstable atomic nucleus. Emitting radiation allows an unstable nucleus to achieve a more stable (lower energy) configuration (often by fragmenting into a smaller nucleus). Although several types of nuclear radiation exist, the three most common types are alpha decay, beta decay, and gamma emission.

In **alpha decay**, an unstable nucleus ejects an **alpha particle** (represented as ${}^4_2\alpha$ or ${}^4_2\text{He}$) consisting of 2 protons and 2 neutrons (a helium-4 nucleus without its electrons), as indicated by the **nuclear notation** of the particle.

Beta decay can occur in more than one form. During **beta minus (β^-)** decay, an **electron** (represented as e^- , ${}^0_{-1}\text{e}$, or β^-) and an antineutrino (represented as $\bar{\nu}_e$ or ${}^0_0\bar{\nu}$) are emitted from the nucleus via the conversion of a neutron to a proton. Alternatively, during **beta plus (β^+)** decay, a **positron** (represented as e^+ , ${}^0_{+1}\text{e}$, or β^+) and a neutrino (represented as ν_e or ${}^0_0\nu$) are emitted from

the nucleus via the conversion of a proton to a neutron through direct transmutation or through electron capture. Although present, the neutrino and antineutrino may not always be explicitly mentioned when discussing beta decay in a chemistry context.

During **gamma emission**, an unstable nucleus in an excited state releases excess energy by emitting a gamma ray (a **high-energy photon**, which is an **electromagnetic wave** packet (represented as γ or ${}^0_0\gamma$), but no particle is emitted and the number of proton and neutrons in the nucleus remains unchanged. As a result, the isotopic and elemental identity of the nucleus are unchanged following gamma emission.

Therefore, an atom of ruthenium-97 undergoing gamma emission would retain its identity and emit a photon.

(Choice A) This reaction shows the β^- decay of ruthenium-97.

(Choice B) This reaction shows the alpha decay of ruthenium-97.

(Choice C) This reaction shows the β^+ decay of ruthenium-97.

Educational objective:

Radioactivity is the spontaneous emission of radiation (photons or particles) from an unstable atomic nucleus. The three most common types of radioactive decay are alpha decay (ejection of an alpha particle), beta decay (ejection of a nuclear electron or positron), and gamma emission (release of a high-energy photon).

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